

DATA MINING

NAME: BALAJI M

REG NO: 192565072



Introduction to Data Mining

Data mining is the process of discovering patterns, correlations, trends, and useful information from large sets of data using techniques from statistics, machine learning, and database systems. It helps organizations transform raw data into meaningful knowledge that supports better decision-making.

Key Techniques

- Classification: Assigning data items to predefined categories or classes.
- Clustering: Grouping similar data points together without predefined labels.
- Association Rule Mining: Finding relationships between variables, such as in market basket analysis.
- Regression: Predicting numerical outcomes based on historical data.

Applications

Data mining is widely used in areas such as market analysis, fraud detection, customer relationship management, healthcare diagnostics, risk management, and recommendation systems. By analyzing large volumes of data, businesses and researchers can uncover hidden patterns that drive innovation and efficiency.

Conclusion

As the volume of data generated continues to grow rapidly, data mining plays an increasingly important role in extracting actionable insights and enabling data-driven decision-making across industries.